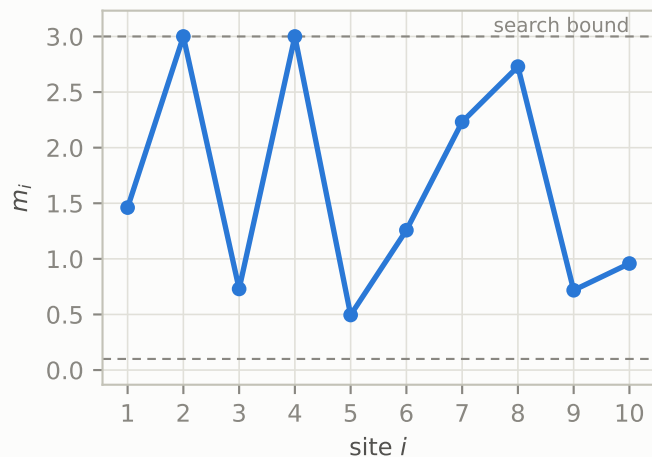


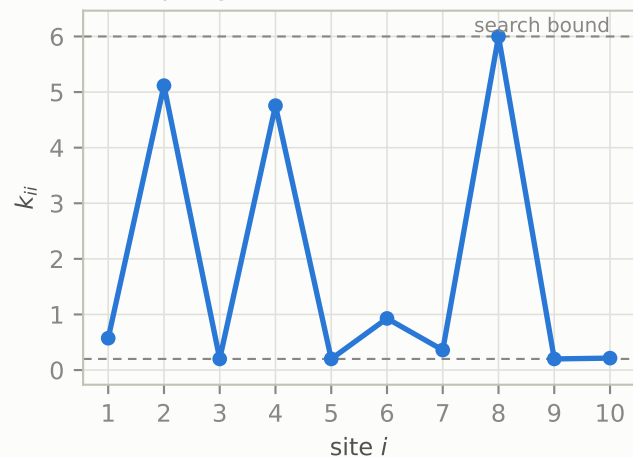
# The route-3 champion: a fully-coupled lab-feasible detector at +21%

Coupling floor 0.02 (all nine bonds active). Worst-mode swept response  $19.11 = 1.208 \times$  the universal uncoupled bound (15.81); all mode gaps  $\geq 3$  linewidths.

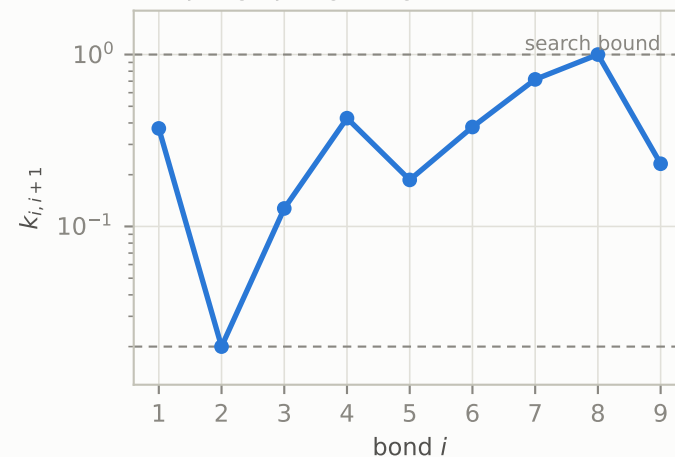
### Masses



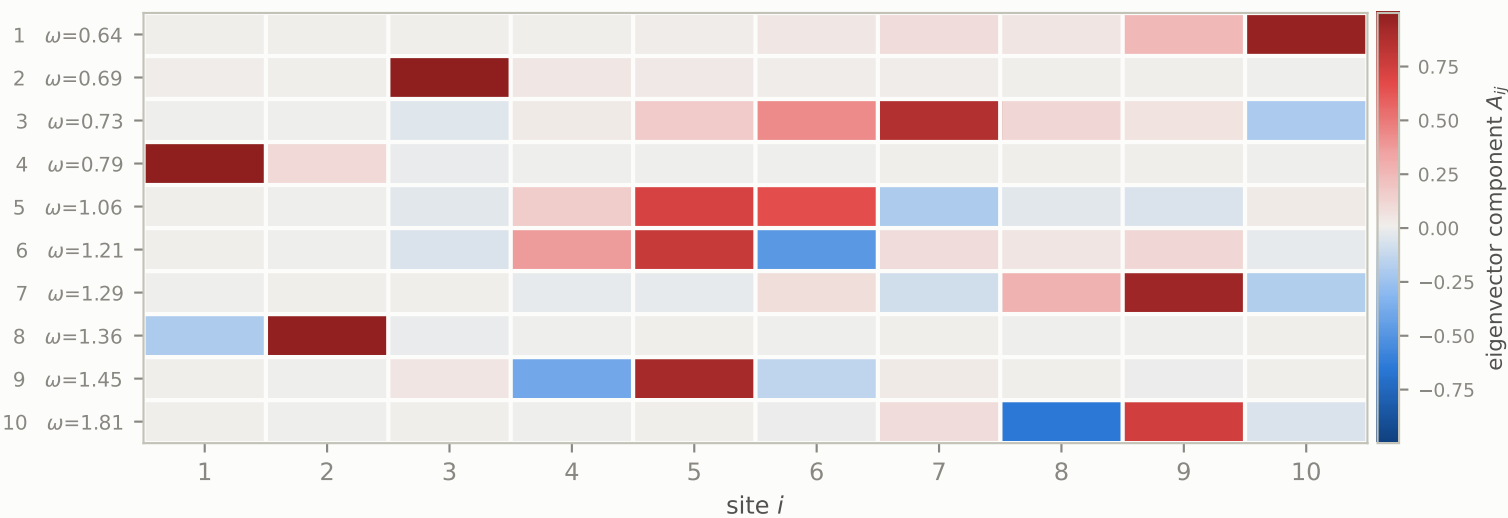
### Wall springs



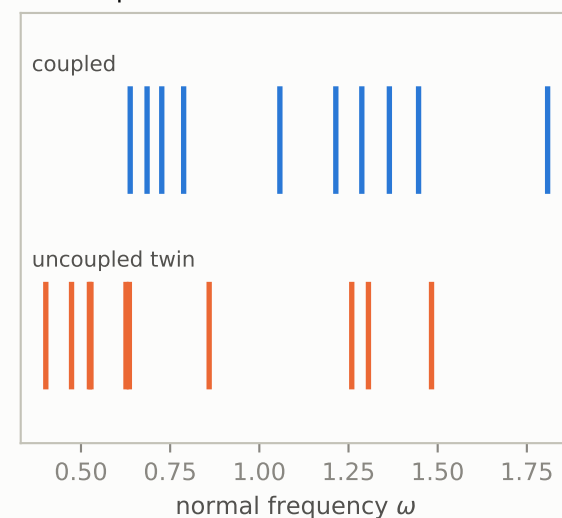
### Coupling springs (log)



### Normal-mode states



### Mode spectra



### Swept response of the optimal observable — every resonance clears the bound

